

Name and Student ID:

Part I (60 points)

1. (Production function)

- a. (True or False) Diminishing marginal return implies diminishing return. (5 points)

F Diminishing marginal return means marginal return will decrease but it's still positive.

Diminishing return means marginal return is negative, so one more input will cause total output to decrease.

- b. If each extra worker produces an extra unit of output, how do the total product of labor, the average product of labor and the marginal product of labor vary with the number of workers? (5 points)

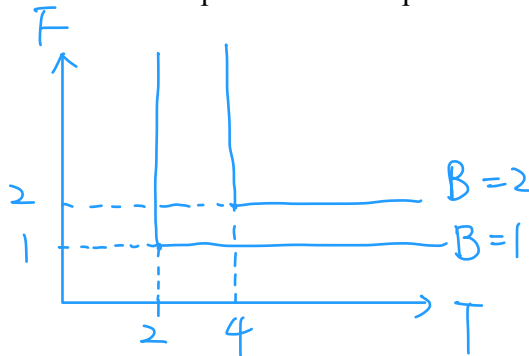
$$y = f(L, \bar{K}) \Rightarrow y = L$$

total product of labor curve is a straight line and the slope of 1.

$$AP_L = \frac{q}{L} = 1$$

$$MP_L = \frac{\partial q}{\partial L} = 1$$

- c. Let B be the number of bicycles produced from F bicycle frames and T tires. Every bicycle needs exactly two tires and one frame. Draw two isoquants for bicycle production and write a mathematical expression for the production function for bicycles. (5 points)



$$\bar{q} = \min\left(\frac{T}{2}, \frac{F}{1}\right)$$

2. (Production function) Suppose the production for widget is given by

$$q = kl - 0.8k^2 - 0.2l^2,$$

where q represents the annual quantity of widgets produced, k represents annual capital input and l represents annual labor input.

- a. Suppose $k=10$. Derive MP_L and AP_L . (6 points)

$$k=10 \Rightarrow q = 10l - 80 - 0.2l^2$$

$$MP_L = \frac{\partial q}{\partial l} = 10 - 0.4l$$

$$AP_L = \frac{q}{l} = 10 - \frac{80}{l} - 0.2l \quad \#$$

- b. Following part a, graph MP_L and AP_L . At what level of labor input does the average productivity reach a maximum? At what level of labor input does MP_L ? (9 points)

when $MP_L = AP_L \Rightarrow AP$ reach maximum

$$10 - 0.4l = 10 - \frac{80}{l} - 0.2l$$

$$\frac{80}{l} = 0.2l$$

$$80 = 0.2l^2$$

$$400 = l^2$$

$$\pm 20 = l \text{ (負不合)}$$

$$\Rightarrow l = 20$$

$$MP_L = 10 - 0.4 \times 20$$

$$= 10 - 8$$

$$= 2$$

$$MP_L = 0 \quad TP_L \text{ maximum}$$

$$10 - 0.4l = 0$$

$$l = 25$$

3. (Production function) Consider a production function whose equation is given by the formula $q = L^a (AK)^b$, where a , b and A are larger than zero.

- a. Determine the elasticity of output to labor. (5 points)

$$\varepsilon_L = \frac{d q / d L}{q/L} = \frac{mP_L}{AP_L} \Rightarrow \frac{a L^{a-1} (AK)^b}{L^{a-1} (AK)^b} = a \quad \#$$

$$mP_L = a L^{a-1} (AK)^b \quad AP_L = L^{a-1} (AK)^b$$

- b. Derive the marginal rate of technical substitution and the elasticity of substitution. (10 points)

$$mRTS = - \frac{mP_L}{mP_K} = - \frac{a L^{a-1} (AK)^b}{L^a b \cdot (AK)^{b-1}} = - \frac{a L^{-1}}{b K^{-1}} = - \frac{a}{b} \times \frac{K}{L}$$

$$\text{elasticity of sub.} = \frac{d(\frac{K}{L})}{\frac{K}{L}} \times \frac{mRTS}{mRTS} = - \frac{b}{a} \times (-\frac{a}{b}) = 1 \quad \#$$

- c. Does the production function exhibit increasing, decreasing, or constant return to scale? (5 points)

$$q = L^a (AK)^b = f(L, K)$$

$$\Rightarrow f(tL, tK) = (tL)^a (tAK)^b = t^a L^a t^b (AK)^b = t^{a+b} (L^a (AK)^b)$$

$$\Rightarrow \text{if } a+b=1 \Rightarrow \text{CRS}$$

$$<1 \Rightarrow \text{DRS}$$

$$>1 \Rightarrow \text{IRS}$$

- d. Is an increase in A more beneficial to the marginal productivity of capital or that of labor? That is, is an increase in A labor-saving or capital-saving technical progress. (5 points)

$$mRTS = - \frac{mP_L}{mP_K} \Rightarrow |mRTS| = \frac{mP_L}{mP_K} \quad mRTS = - \frac{a}{b} \times \frac{K}{L} \quad \text{neutral technical progress}$$

$mRTS$ doesn't depend on A

$$mP_K = L^a \cdot b (AK)^{b-1}$$

$|mRTS| \downarrow$

if $b-1 > 0 \Rightarrow$ increase in $A \rightarrow$ increase $mP_K \Rightarrow$ labor-saving technical progress

$b-1 < 0 \Rightarrow$ increase in $A \rightarrow$ decrease mP_K

$\Rightarrow |mRTS| \uparrow$
capital-saving technical progress

Part II: (60 points)

1. (Labor Supply) Rachel maximizes her utility $U(C, N)$ over leisure (N) and consumption (C) subject to the following constraints:

$$H = 24 - N \text{ and}$$

$$C = wH,$$

where H is the hours of work per day and w is the wage rate.

- a. What are the marginal cost and marginal benefits of leisure? **Use your intuition** to derive the first order condition for Rachael's optimal labor supply. (10 points)

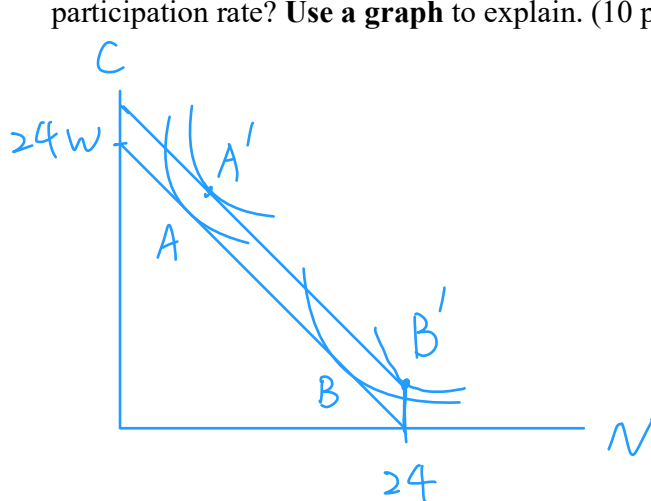
if spend one unit in leisure

$$\Rightarrow mB = mU_N$$

$$mC = w mU_C$$

$$\Rightarrow mB = mC \Rightarrow mU_N = w mU_C$$

- b. What are the effects of an unconditional income transfer on labor supply and labor force participation rate? **Use a graph** to explain. (10 points)



$$H = 24 - N \quad C = wH$$

$$\Rightarrow N = 24 - H$$

$$N = 24 - \frac{C}{w}$$

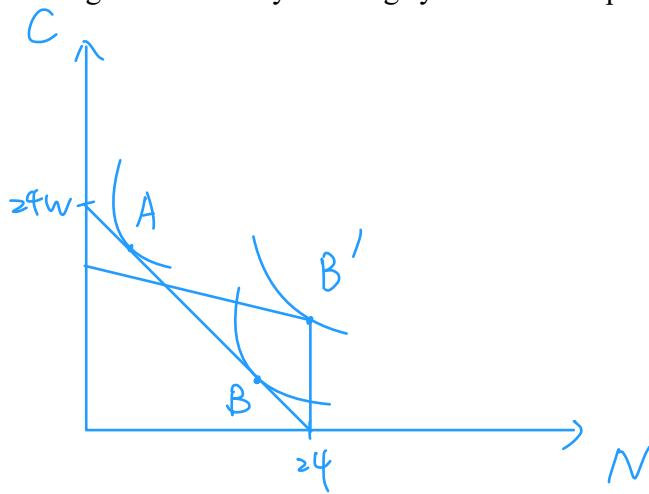
A: hours of work ↓

B: drop out of labor force

$\Rightarrow$ hours of work ↓

labor participation rate ↓

- c. Following part b, suppose the unconditional income transfer is financed by a tax rate, t , on the wage rate. Would you change your answer in part b? **Use a graph** to explain. (10 points)



B: low wage worker

$\Rightarrow$ hours of work $\downarrow$ labor participation rate $\downarrow$

A: high wage worker

SE: $N \uparrow$ $\Rightarrow$ we can't have a clear understanding of
 IE: $N \downarrow$ A, hours of work?

labor participation rate will drop, but the effect
 on hour of work is uncertain

2. (Two-Period Consumption Model) Carol has income y_1 in period 1 and y_2 in period 2 to allocate between consumption in two periods (c_1 and c_2). Her utility function is given by

$$u(c_1, c_2) = \ln c_1 + \beta \ln(c_2),$$

and the budget constraint is

$$y_1 + \frac{y_2}{1+r} = c_1 + \frac{c_2}{1+r}$$

where r is the one-period interest rate.

$$(1+r)y_1 + y_2 = \frac{C_1(1+r)}{\frac{1+\beta}{\beta}} + C_2$$

- a. Find her optimal choice for c_1 and c_2 . Derive $\frac{\partial c_1}{\partial r}$ and $\frac{\partial c_2}{\partial r}$. (10 points)

$$MRS_{1,2} = -\frac{MU_1}{MU_2} = -(1+r) \Rightarrow \frac{1/c_1}{\beta/c_2} = 1+r \Rightarrow C_2 = \beta C_1 (1+r)$$

$$C_1^* = \frac{1}{1+\beta} \left(y_1 + \frac{y_2}{1+r} \right) \quad C_2^* = \frac{\beta}{1+\beta} \left(y_1 + \frac{y_2}{1+r} \right)$$

$$(1+t_1)C_1 + \frac{(1+t_2)\beta(1+r)C_1}{1+r} C_1$$

- b. Suppose the government imposes a tax rate, t_1 , on the consumption in period 1 and a tax rate, t_2 , on the consumption in period 2. Discuss the effects of a decrease in t_1 on savings and the consumptions in two periods. (10 points)

$$C_2 = \beta(1+r)C_1$$

$$(1-t_1)C_1 + \frac{(1-t_2)C_2}{1+r} = y_1 + \frac{y_2}{1+r}$$

$$C_1^* = \frac{y_1 + \frac{y_2}{1+r}}{1-t_1 + (1-t_2)\beta}$$

$$\text{let } X = y_1 + \frac{y_2}{1+r} \quad T = (1-t_1) + (1-t_2)\beta \quad C_2^* = \beta(1+r) \frac{y_1 + \frac{y_2}{1+r}}{1-t_1 + (1-t_2)\beta}$$

$$C_1^* = \frac{X}{T} \quad C_2^* = \beta(1+r) \frac{X}{T}$$

$$\frac{\partial C_1^*}{\partial t_1} = \frac{-X}{T^2} \quad \frac{\partial C_2^*}{\partial t_1} = \beta(1+r) \frac{-X}{T^2} \quad \frac{\partial S_1^*}{\partial t_1} = -\frac{\partial C_1^*}{\partial t_1} \quad \frac{\partial S_1^*}{\partial t_1} = -\frac{\partial C_1^*}{\partial t_1}$$

$$\Rightarrow t_1 \downarrow \rightarrow C_1 \uparrow, C_2 \uparrow \text{ but } S_1 \downarrow$$

- c. Following part b, suppose the t_1 and t_2 are equal initially. Discuss the effects of a permanent tax rate decrease on savings and the consumptions in two periods. (10 points)

$$\frac{\partial C_1^*}{\partial t_1} + \frac{\partial C_2^*}{\partial t_2} = \frac{-X}{T^2} - \frac{\beta X}{T^2} = - \frac{(1+\beta)X}{T^2}$$

$$\frac{\partial C_1^*}{\partial t_2} + \frac{\partial C_2^*}{\partial t_2} = - \frac{\beta(1+r)X}{T^2} - \frac{\beta^2(1+r)X}{T^2} = - \frac{(1+r)(\beta+\beta^2)X}{T^2}$$

$$\frac{\partial S_1^*}{\partial t_1} + \frac{\partial S_1^*}{\partial t_2} = \frac{(1+\beta)X}{T^2}$$

A permanent decrease in t

$\Rightarrow C_1 \uparrow, C_2 \uparrow, S_1 \downarrow$

with larger magnitude than part a